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Roll.No :- 2401001 **Sem – V** **Batch : T11** **LO Mapped : LO - 03**

EXPERIMENT – 06

Aim : To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

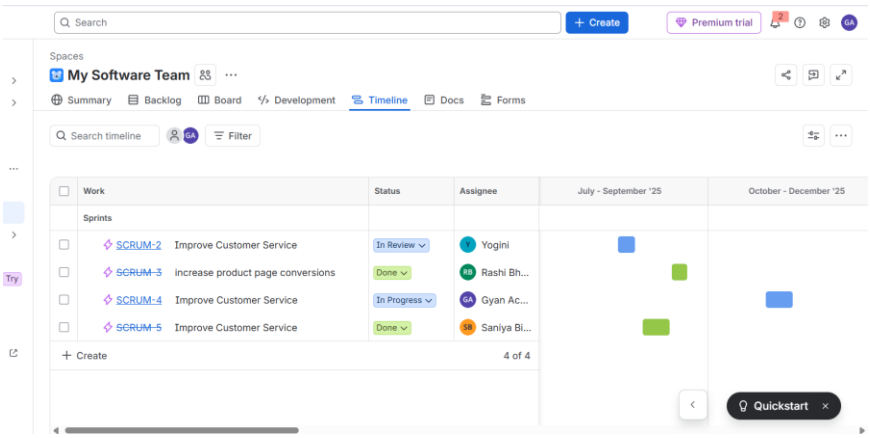
Theory : Agile Software Development is an iterative and incremental approach to software development in which a project is divided into small, manageable tasks and developed through repeated cycles called iterations or sprints. Unlike traditional software development models, Agile focuses on continuous customer feedback, frequent delivery of working software, collaboration among team members, and the ability to respond quickly to changing requirements. The Agile lifecycle generally consists of requirement gathering, product backlog creation, sprint planning, development, testing, review, and retrospective. Each sprint produces a potentially usable increment of the software, allowing the development team to continuously improve the product.

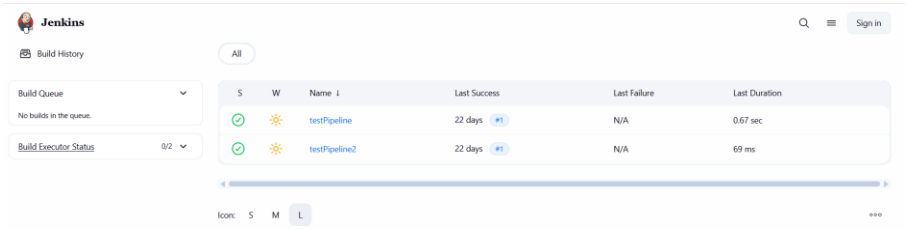
Jira is a project management and issue-tracking tool widely used to implement Agile methodologies such as Scrum and Kanban. In Jira, project requirements are represented as issues such as epics, user stories, tasks, and bugs. An Epic represents a large feature or requirement, while a User Story describes a functionality from the user's perspective. Tasks represent specific activities required to complete the work, and bugs are used to record defects found during development or testing. Jira provides features such as product backlogs, sprint boards, Kanban boards, workflows, reports, dashboards, and issue tracking, which help teams organize and monitor the complete software development lifecycle.

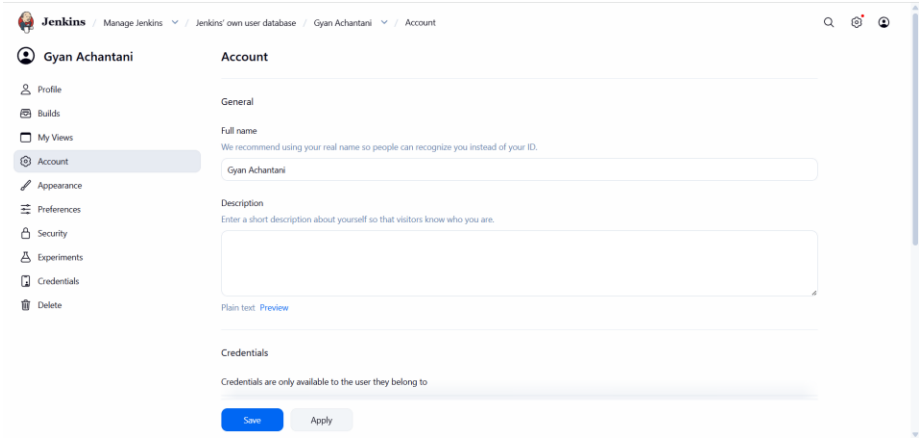
The Agile project lifecycle using Jira begins with identifying project requirements and creating the product backlog. The product owner prioritizes the backlog according to business value and project requirements. During Sprint Planning, the development team selects suitable backlog items for the upcoming sprint and divides them into smaller tasks. Developers then implement the selected tasks, while testers verify the functionality and report defects in Jira. The progress of

each issue can be tracked using statuses such as To Do, In Progress, Code Review, Testing, and Done. At the end of the sprint, the team conducts a Sprint Review to demonstrate the completed work and a Sprint Retrospective to identify improvements for future sprints.

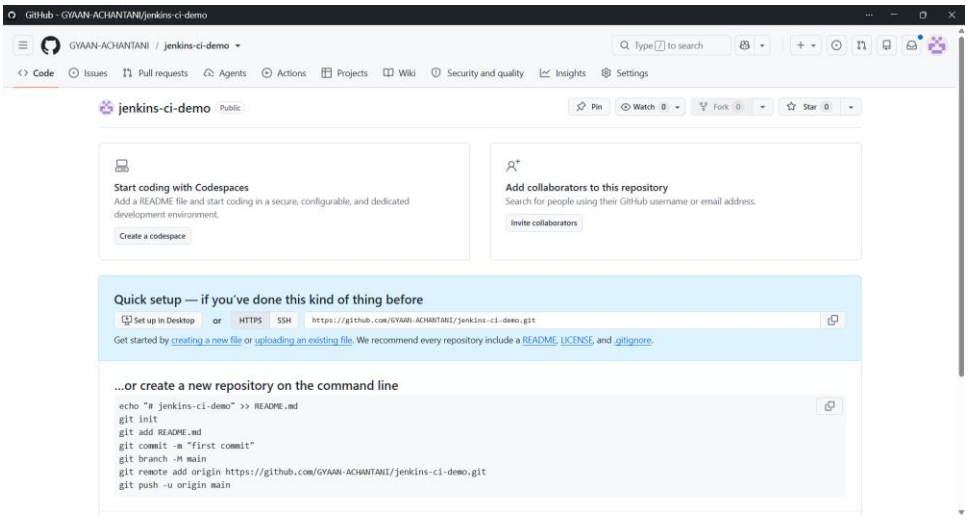
Implementation :

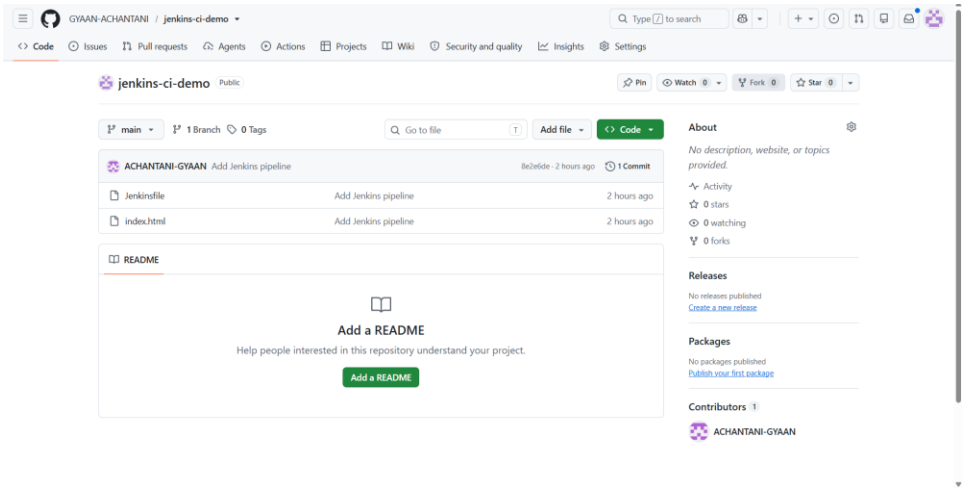






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```
C:\Users\Gyaan\OneDrive\Documents\jenkins-ci-demo>git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 252 bytes | 252.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/GYAAN-ACHANTANI/jenkins-ci-demo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Conclusion :- The experiment successfully demonstrated the Agile Software Project Lifecycle using Jira with DevOps Pipeline Integration. Jira was used to manage requirements, user stories, tasks, sprints, and bugs, while the DevOps

pipeline provided a systematic process for code integration, automated testing, and deployment. The integration improved project visibility, collaboration, traceability, and continuous delivery. Thus, combining Agile practices with Jira and DevOps provides an efficient, automated, and reliable approach for managing and delivering modern software projects.